

Kali Linux Live USB — Complete Guide

OS: Kali Linux (Live) · **Upstream:** kali.org (Debian-based, Offensive Security) **Type:** Live amd64 ISO written whole-disk to a USB · **Cyber Controller catalog id:** `kali` (Software-OS tab, removable-USB writer; `verify_model = checksums_sig`) **This guide:** choose a USB → verify → flash to USB via Cyber Controller's Software-OS tab → boot → add (encrypted) persistence → troubleshoot UEFI/Secure Boot.

1. Overview

Kali Linux is a Debian-based **penetration-testing / security** distribution. The **Live** `amd64` image boots straight from a USB stick into a full Kali desktop **without installing anything or touching the machine's internal disk** — "a full bare metal Kali install without needing to alter an already-installed operating system" (kali.org, §9). It ships the Kali toolset (network/recon, wireless, web, exploitation and forensics tooling). It is *not* amnesic like Tails and it is *not* an installer like Arch — it is a portable, optionally-persistent working environment. Cyber Controller's **Software-OS tab** downloads the current Live image from the official mirror, integrity-verifies it, and writes it to a confirmed removable USB. This is the OS/SD-writer path; it is **separate from the firmware Flash tab** (`esptool`) used for ESP32 boards.

2. Legal & Safety

Authorized testing only. Kali bundles offensive tooling (scanners, exploit frameworks, wireless attack tools). Running these against networks, systems, or people you do not own or lack **written permission** to test is illegal in most jurisdictions (US: CFAA; EU and others similar). Kali itself is a lawful, freely distributed operating system — possessing and learning it is legal; *unauthorized use* is not. Use only your own lab gear or an explicit engagement scope. Two data-safety notes: (a) flashing **erases the entire target USB**, and (b) Kali Live by default writes nothing to the host's internal disk, but installer/persistence steps can — double-check the target device before you write.

3. Hardware & Purchasing

You only need a **USB flash drive (8GB+; Kali Live with persistence wants 16GB+)** and a 64-bit (`amd64/x86-64`) PC to boot it.

Item	Recommendation	Notes / search terms
USB stick	8GB minimum , 16–32GB recommended	The Live ISO is >4GB, so 8GB barely fits the OS alone; a persistence partition needs the rest (kali.org persistence docs, §9). Search: "USB 3.2 flash drive 32GB"

Speed	USB 3.x flash drive (or USB-SSD)	Live runs from the stick — disk ops "may slow due to the utilized storage media" (kali.org, §9). A fast drive transforms the experience.
Endurance (optional)	Small USB-SSD in an enclosure	If you'll use persistence heavily, an SSD outlasts cheap flash and is much faster.
Host PC	Any x86-64 PC <~10 yrs old	Needs a USB-bootable BIOS/UEFI.

Buy from any reputable retailer (Amazon, Best Buy, Micro Center, manufacturer stores — SanDisk, Samsung, Kingston). There is no special hardware to purchase; **do not** download Kali from third-party "mirrors" you find via ads — use the official site (§9). *Verify: counterfeit/oversized fake-capacity USB sticks exist; test capacity (e.g. `f3/h2testw`) on suspiciously cheap drives.*

4. Making the bootable USB / partitioning notes

Cyber Controller writes the Live ISO **raw (block-for-block)** to the whole USB device — this is an "isohybrid" image, so the resulting stick is bootable on both UEFI and legacy BIOS. Key facts:

- **The whole drive is overwritten.** Any existing partitions/data on the USB are destroyed. The image's own partition table replaces whatever was there.
- **ISO, written directly to the device** (not "burned" as files, not copied to a FAT32 partition). Tools like Rufus's "DD mode", `dd`, or Cyber Controller's writer all do the same raw write.
- **Persistence is a partition you add after writing the image** (see §7). Plan capacity now: image (>4GB) + your desired persistence space $\leq$ USB size, so pick 16GB+ if you want meaningful persistence.
- **Ventoy alternative:** instead of a dedicated single-OS stick, you can install **Ventoy** to a USB and then just *copy* the Kali `.iso` onto it (multi-ISO boot menu). That is a separate workflow from Cyber Controller's raw writer — convenient for carrying several ISOs, but it does not set up Kali persistence the same way. Search: "Ventoy multiboot USB".

5. Flash to USB via Cyber Controller's Software-OS tab

1. Insert the USB. Open Cyber Controller → **Software-OS** tab (labeled "Write a verified bootable operating system to a USB stick").
2. **Operating System:** pick **Kali Linux (Live)**. The description shows it is an ISO verified via `checksums_sig`.
3. **Check latest:** click it. Cyber Controller resolves the current release **live** by parsing `cdimage.kali.org/current/SHA256SUMS` (the `current` path always points at the newest release) and reads the image filename + SHA-256 from that file. If you have no network, tick **"Use bundled version (offline)"** to flash the version pinned in the catalog (offline fallback; at time of writing the pinned fallback is `2026.2` — the live check supersedes it).
4. **Target USB (removable):** select your stick. **Only removable drives are listed** (the writer refuses fixed/system disks and anything ≥ 256 GB). Click **Refresh drives** if it's missing.
5. **(Pro mode) Use local image...** optionally point at a Kali `.iso` you already downloaded instead of fetching it. In **Simple mode** this and the offline toggle are hidden and it always fetches the latest.

6. Click **Flash OS** → confirm the "ERASE EVERYTHING" dialog. Cyber Controller downloads (if needed), **verifies** (see §6), writes the image, then **reads it back and re-hashes** to confirm the write.
7. On success: *"boot the target machine from this USB."*

CLI equivalent: `cyber-controller --flash-os kali [--os-image <local.iso>] [--os-target <device>] [--offline] [--yes]`. The flash refuses to run without confirmation (the whole USB is erased).

6. Integration with the Software-OS tab (integrity verification)

- **Catalog entry** (`src/config/os_catalog.json`): `id kali, resolver: kali, image_type: iso, verify_model: checksums_sig, kali_variant: live-amd64`, official GPG fingerprint `827C 8569 F251 8CC6 77FE CA1A ED65 462E C8D5 E4C5`.
- **Live resolve:** the kali resolver fetches SHA256SUMS from `cdimage.kali.org/current/`, finds the `kali-linux-<ver>-live-amd64.iso` line, and takes its filename, version, and SHA-256. All OS downloads are restricted to an allowlist of official hosts (`cdimage.kali.org`, `kali.download`), HTTPS-only, with redirects re-validated (anti-SSRF).
- **Verification model — checksums_sig (this is why Kali is a two-step verify):** 1. Cyber Controller downloads SHA256SUMS **and** SHA256SUMS.gpg, then verifies the OpenPGP signature on the SHA256SUMS file against the pinned Kali key fingerprint above. *A bad/foreign signature → the flash is refused.* 2. It then hashes the ISO and confirms that SHA-256 **appears in the signed SHA256SUMS**. Mismatch → refused. This chains trust: signed list → list contains image hash → image matches.
- If gpg isn't installed, the signature step is skipped with a NOTE and the SHA-256 check still runs; you'll see a warning that you should verify the signature for full assurance. An image with **no** valid signature *and* no checksum is written only after an explicit UNVERIFIED warning — avoid that.
- **Manual cross-check (recommended):** import the key and verify yourself (`kali.org`, §9): `gpg --keyserver htps://keyserver.ubuntu.com --recv-key 827C8569F2518CC677FECA1AED65462EC8D5E4C5` then `gpg --verify SHA256SUMS.gpg SHA256SUMS` — look for "Good signature from 'Kali Linux Archive Automatic Signing Key'" with that key ID.
- **Post-write read-back:** after writing, the backend re-reads the bytes from the USB and compares the SHA-256 to the image — a hardware-level confirmation the stick is good.

7. Booting it / persistence

Boot: insert the stick, power on, open the firmware **boot menu** (commonly **F12 / F10 / F9 / Esc**, or **Option** on Intel Macs — *verify your model*), and pick the USB. At the Kali boot menu choose **Live (amd64)**. *Default live credentials are `kali / kali` — verify on the current Kali Live boot screen, as defaults have changed across releases.*

Persistence (optional, makes the stick remember changes): - Requires spare space on the USB (so the **8GB minimum / 16GB+ recommended**). After writing the image, add a new partition in the free space, format it **ext4** and **label it persistence**, then create a file `persistence.conf` on it containing the single line `/ union` (`kali.org persistence docs`, §9): `sudo mkfs.ext4 -L persistence /dev/sdX3` then `echo "/ union" | sudo tee /mnt/usb/persistence.conf`. - At boot, pick the **"Live USB Persistence"** menu entry to load your saved changes. - **Encrypted persistence (LUKS):** Kali documents an *"Encrypted Persistence"* variant that puts the persistence partition inside a LUKS container so your saved data is protected at rest — strongly preferred if the stick holds engagement data. Follow Kali's dedicated encrypted-persistence guide for the `cryptsetup luksFormat` steps (§9). - You can keep **multiple labeled stores** (e.g. `work`, `ctf`) and

choose which to load via the `persistence-label=` boot parameter (edit the entry with Tab/e).

8. Troubleshooting (boot / UEFI / Secure Boot)

- **USB not in the boot menu:** enable USB boot in firmware; try a different port (rear/USB-2 port); re-seat the stick. Some firmwares hide USB until **Fast Boot** is disabled.
- **Secure Boot:** current Kali Live is signed for Secure Boot and generally boots with it on; if it refuses, **disable Secure Boot** in firmware (and re-enable after). *Verify against the Kali release notes for your version.*
- **UEFI vs Legacy/CSM:** if it won't boot, toggle between **UEFI** and **Legacy/CSM** boot modes — the isohybrid image supports both, but the firmware must be pointed at the matching one.
- **Black screen / hangs at boot:** at the boot menu press e/Tab and add `nomodeset` (graphics driver fallback) to the kernel line. *Verify the exact parameter for your GPU.*
- **Boots but feels slow:** that's expected on slow flash media (kali.org, §9) — use a USB 3.x drive or add persistence on faster media.
- **Persistence not loading:** confirm the partition label is exactly `persistence`, the file is `persistence.conf` containing `/ union`, and you selected the **Persistence** boot entry.
- **Flash failed / not removable:** the writer only lists removable drives <256GB — if your stick doesn't appear, it may be reported as fixed; try another stick/reader. Run the app **as Administrator** (Windows) or with the needed privileges (Linux/macOS) for raw disk access.
- **Verify failed (checksum/signature):** re-download — a corrupted download or a tampered mirror is the usual cause. Never write an image that fails verification.

9. Sources

- Get Kali (live image): <https://www.kali.org/get-kali/>
- Kali USB docs (writing, persistence, verification index): <https://www.kali.org/docs/usb/>
- Kali USB persistence (partition + `persistence.conf`): <https://www.kali.org/docs/usb/usb-persistence/>
- Verify Kali images securely (SHA256SUMS, `.gpg`, key fingerprint): <https://www.kali.org/docs/introduction/download-images-securely/>
- Cyber Controller catalog + writer: `src/config/os_catalog.json` (entry `kali`), `src/core/os_catalog.py` (resolver/verify), `src/core/backends/sd_backend.py` (removable-only write + read-back), `src/ui/qt/software_tab.py` (Software-OS tab).
- Verify the current Live version, file name, and credentials at flash time — releases roll forward and the tab resolves the latest live.